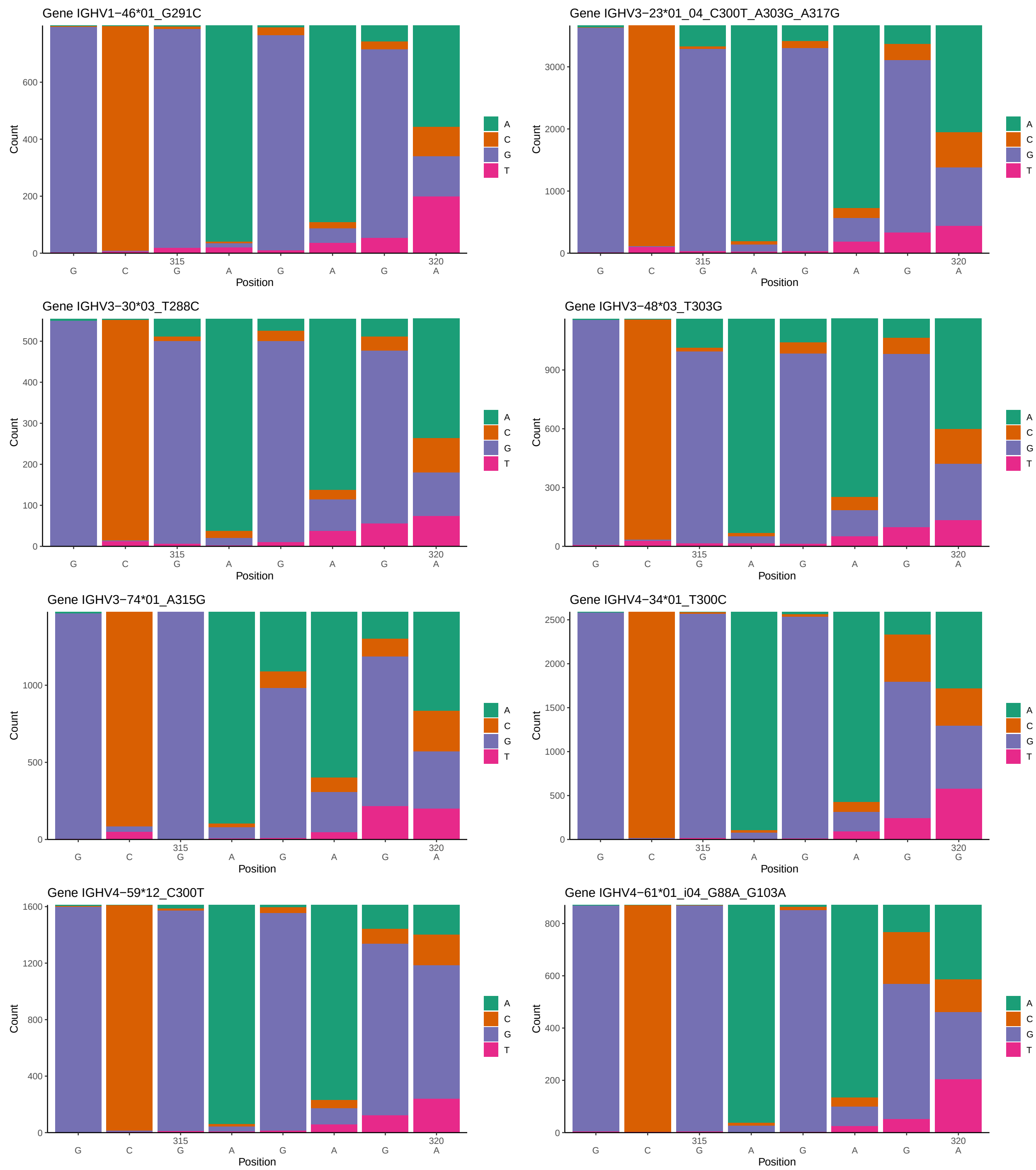
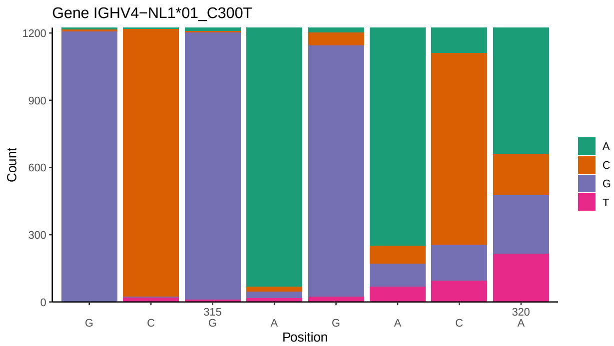
# OGRDBstats Report

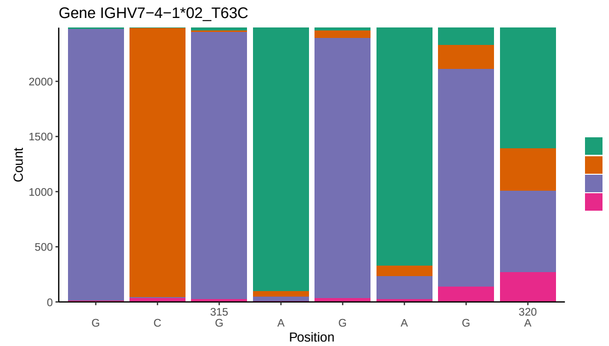
## Contents

# 1 Novel sequence analysis

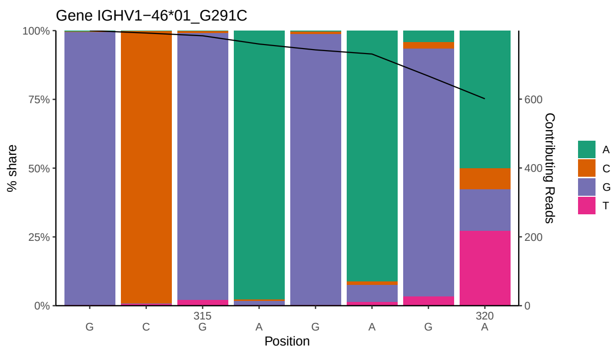
## 1.1 End-nucleotide composition

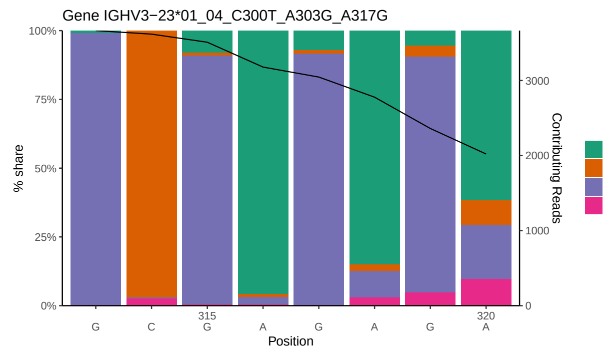


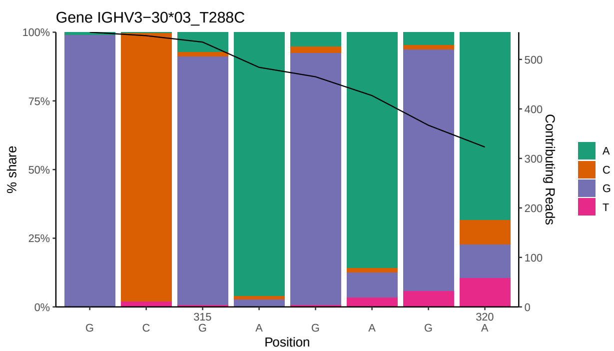


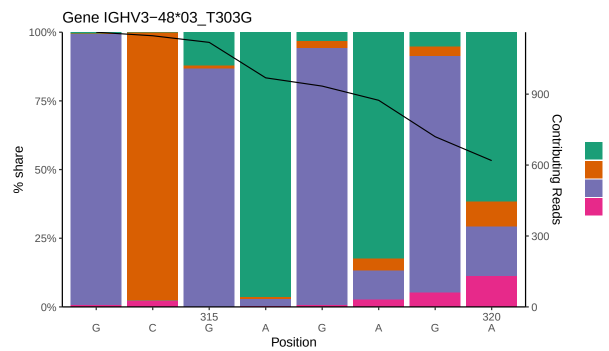



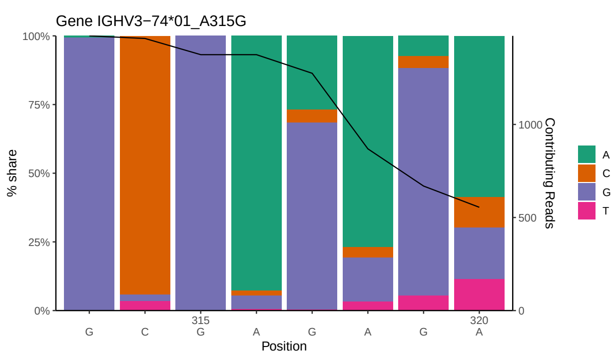

## 1.2 Per-nucleotide consensus where previous nucleotides match the consensus

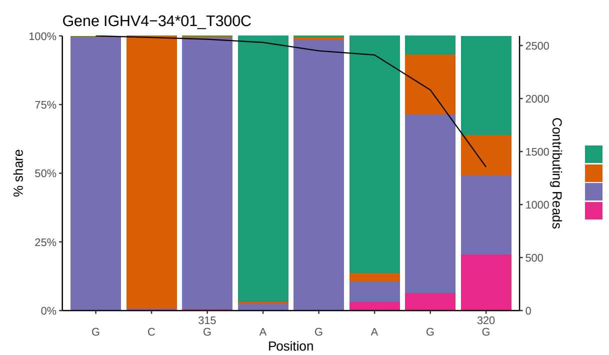


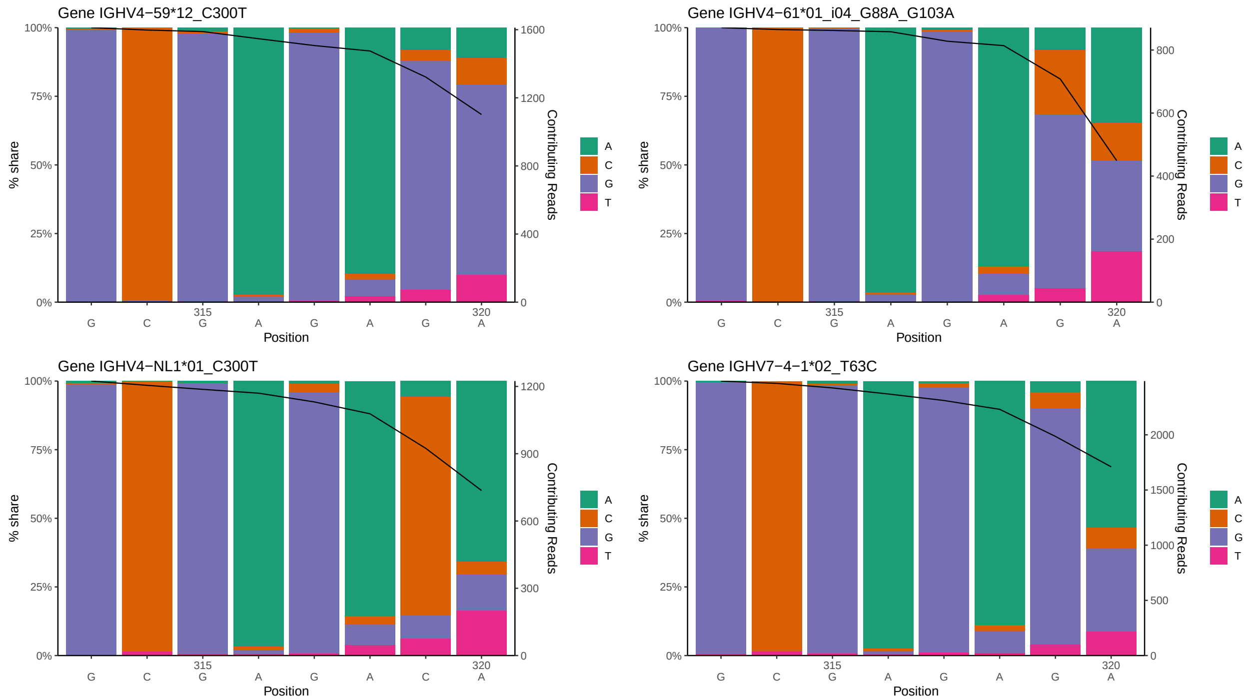



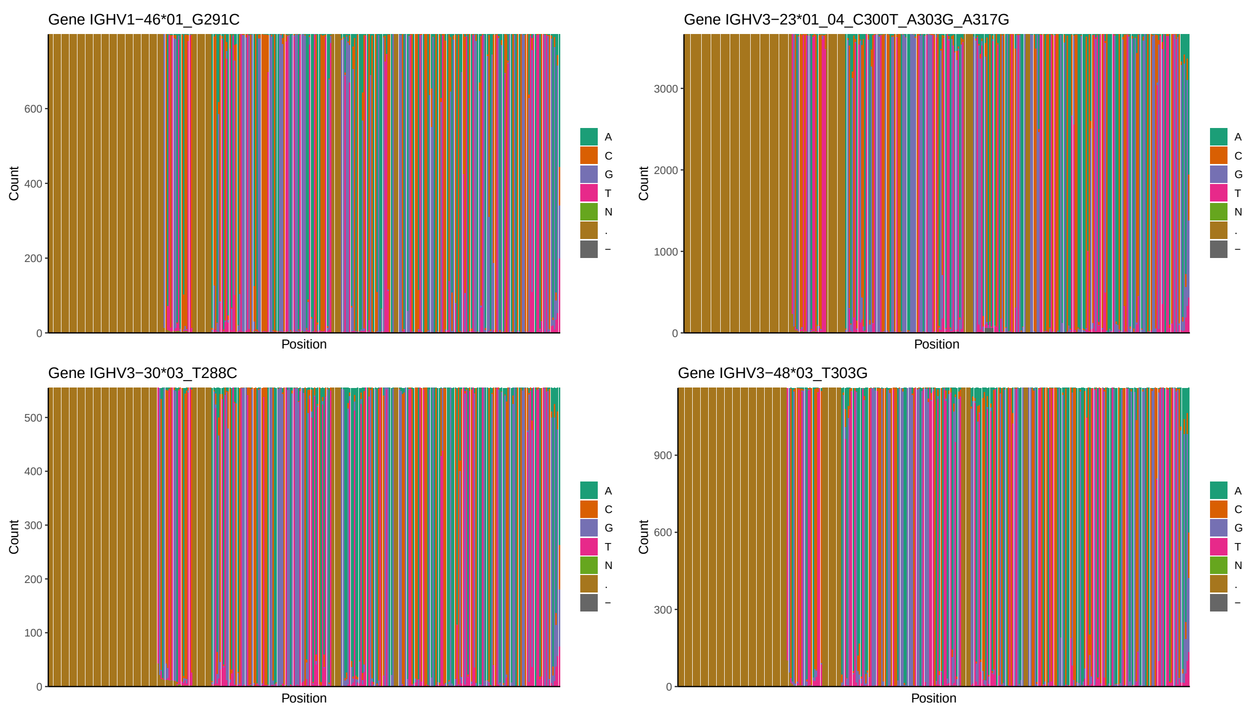



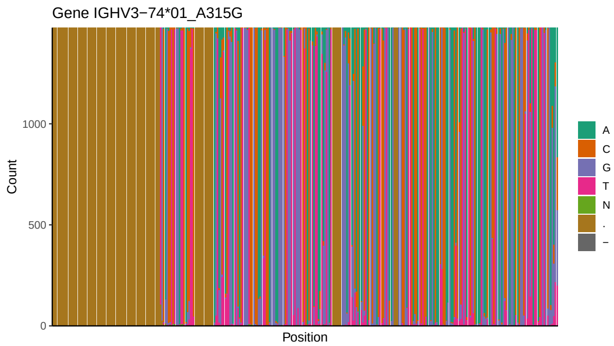



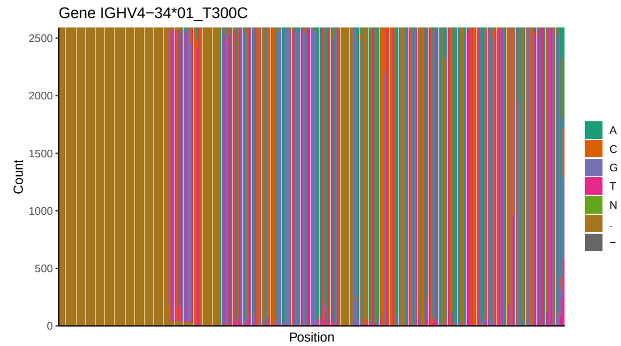



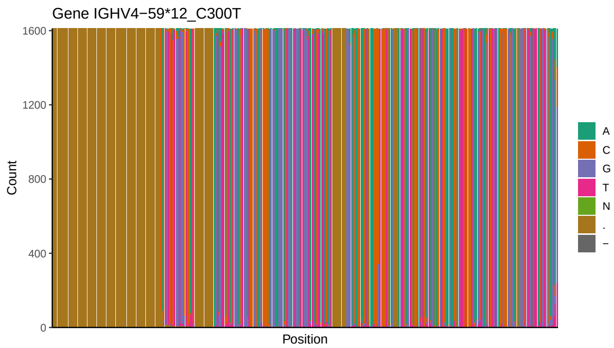


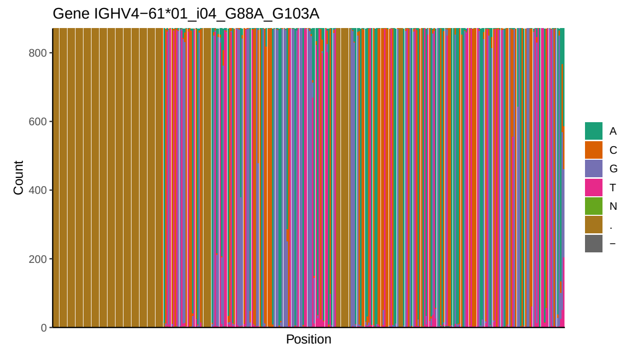

## 1.3 Whole-sequence composition of each assigned read

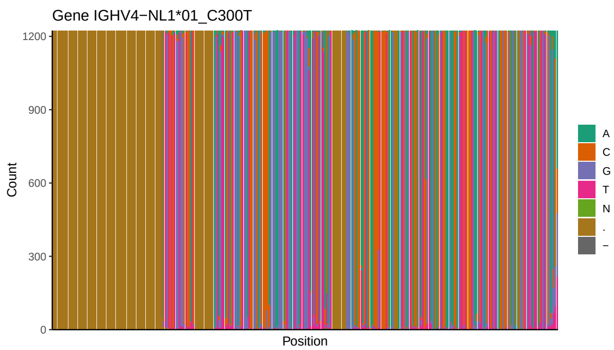


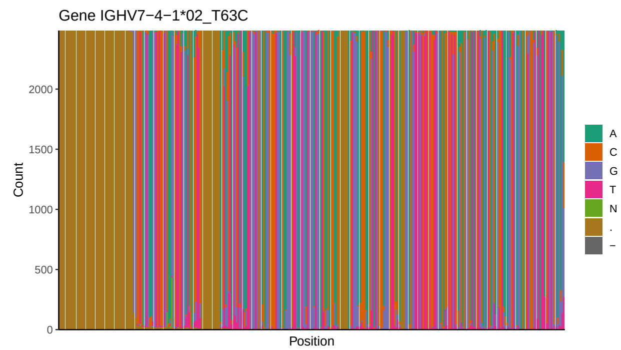


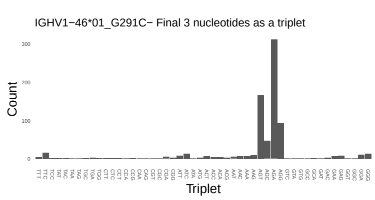



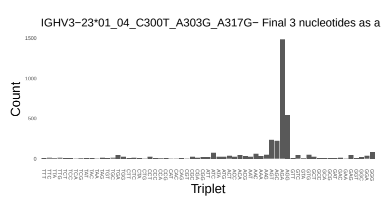



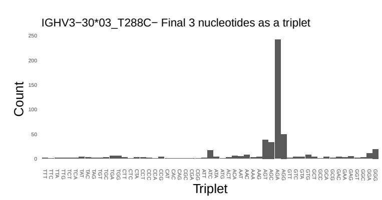



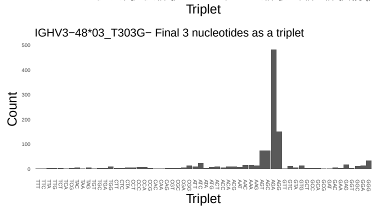



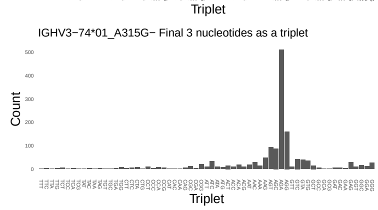

## 1.4 Final three nucleotides: frequency of each observed triplet

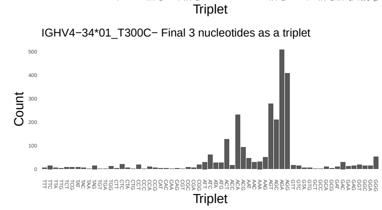


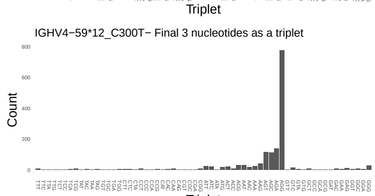



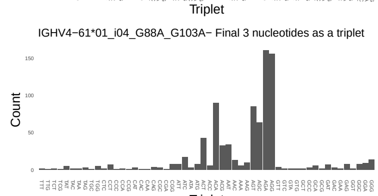



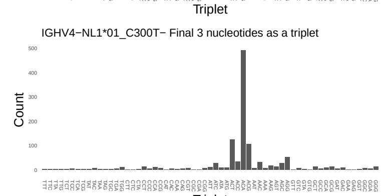



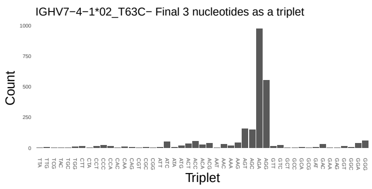



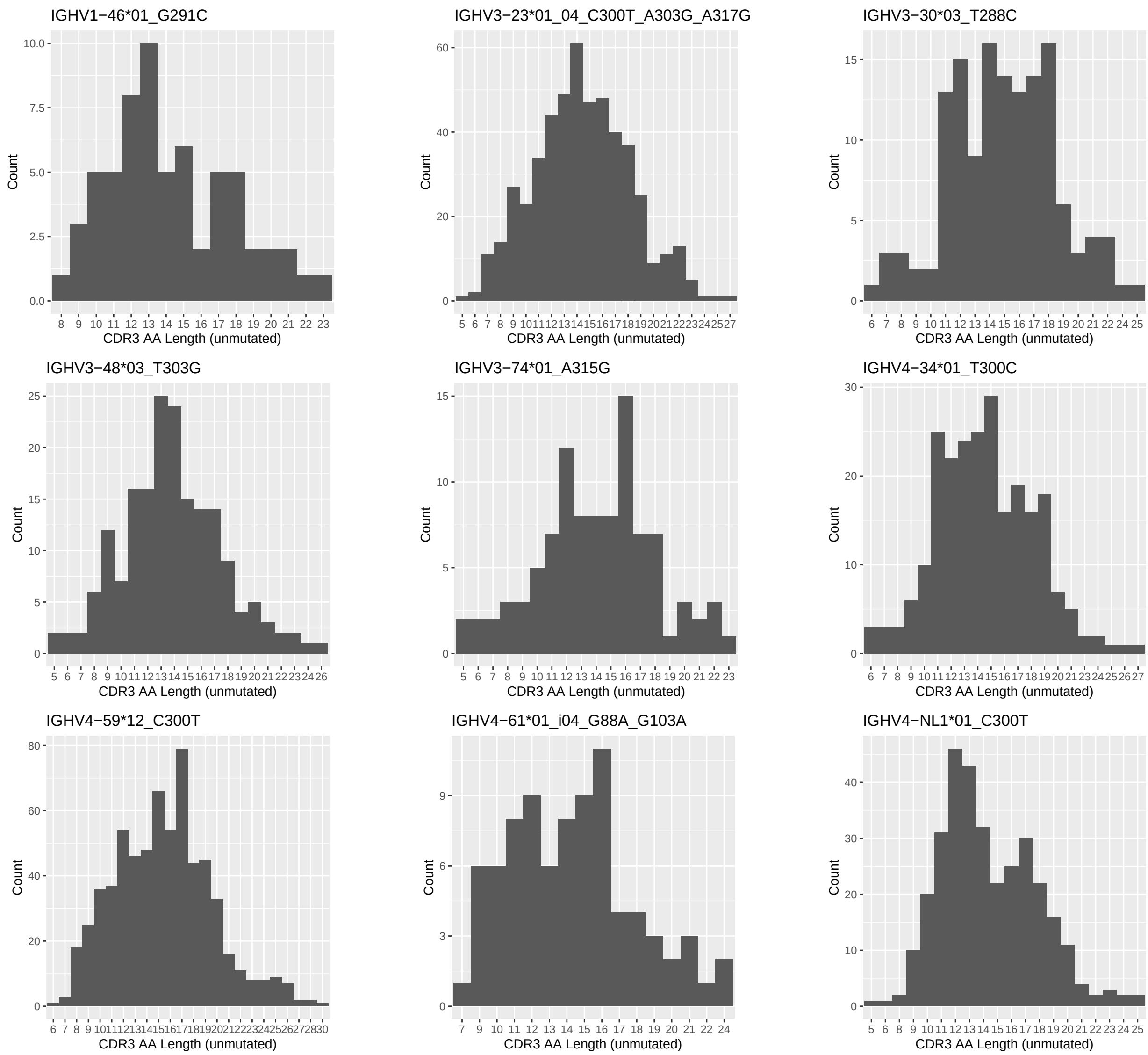



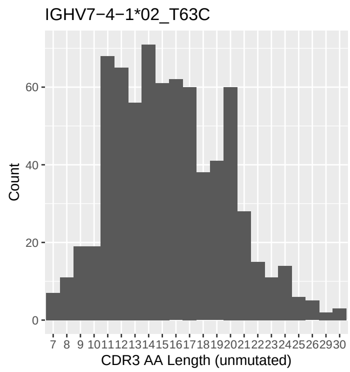



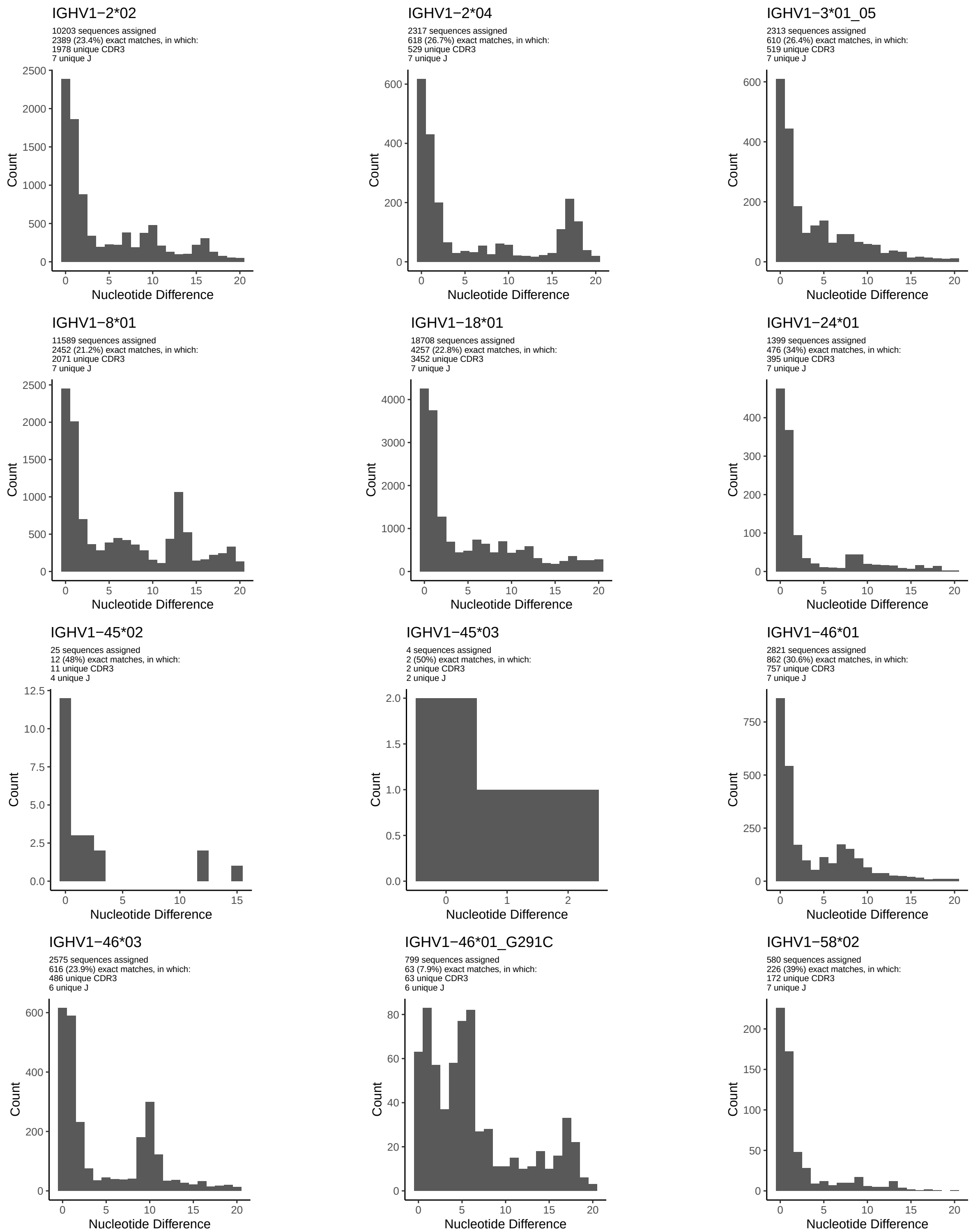


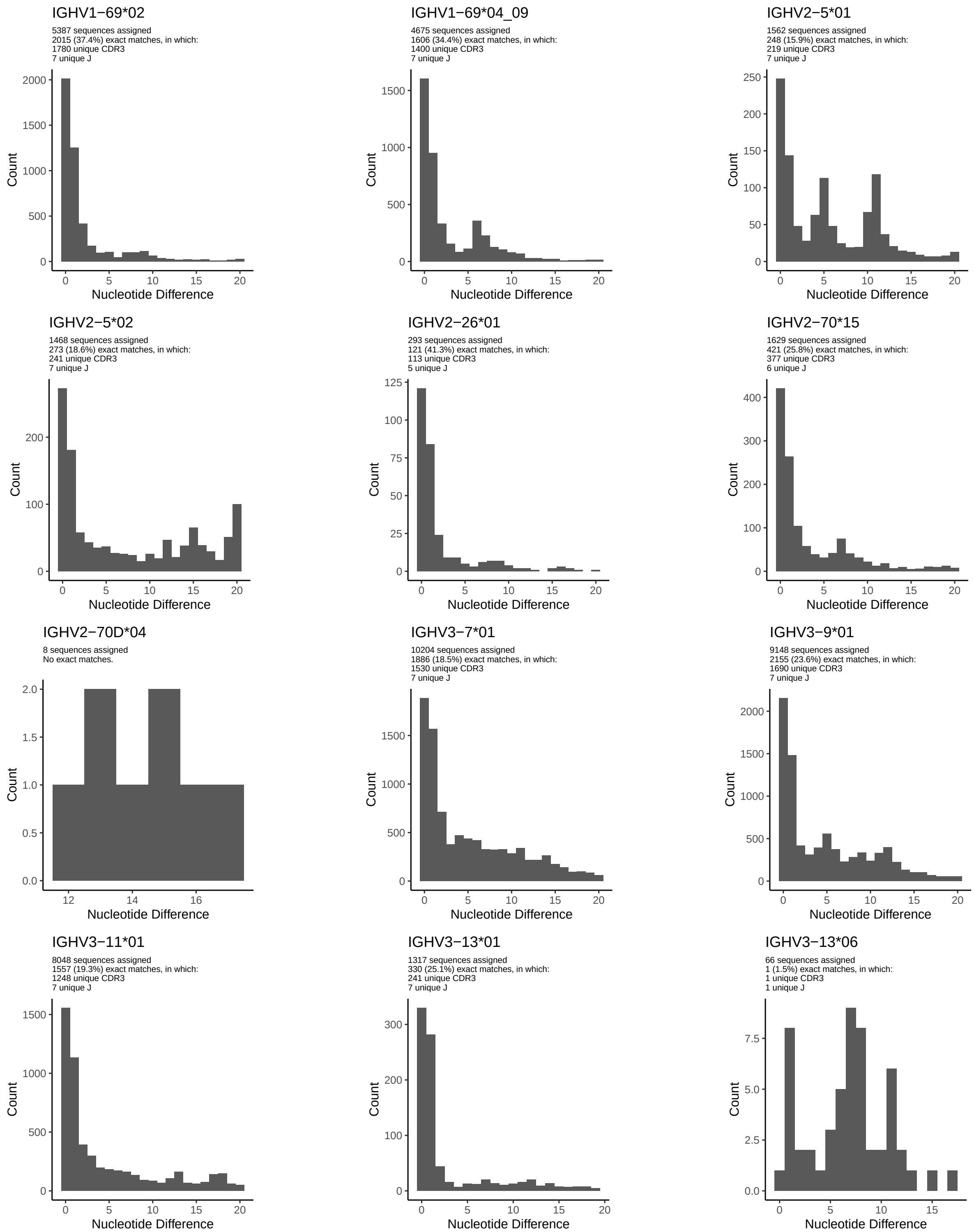

## 1.5 CDR3 length distribution, in assignments to novel alleles

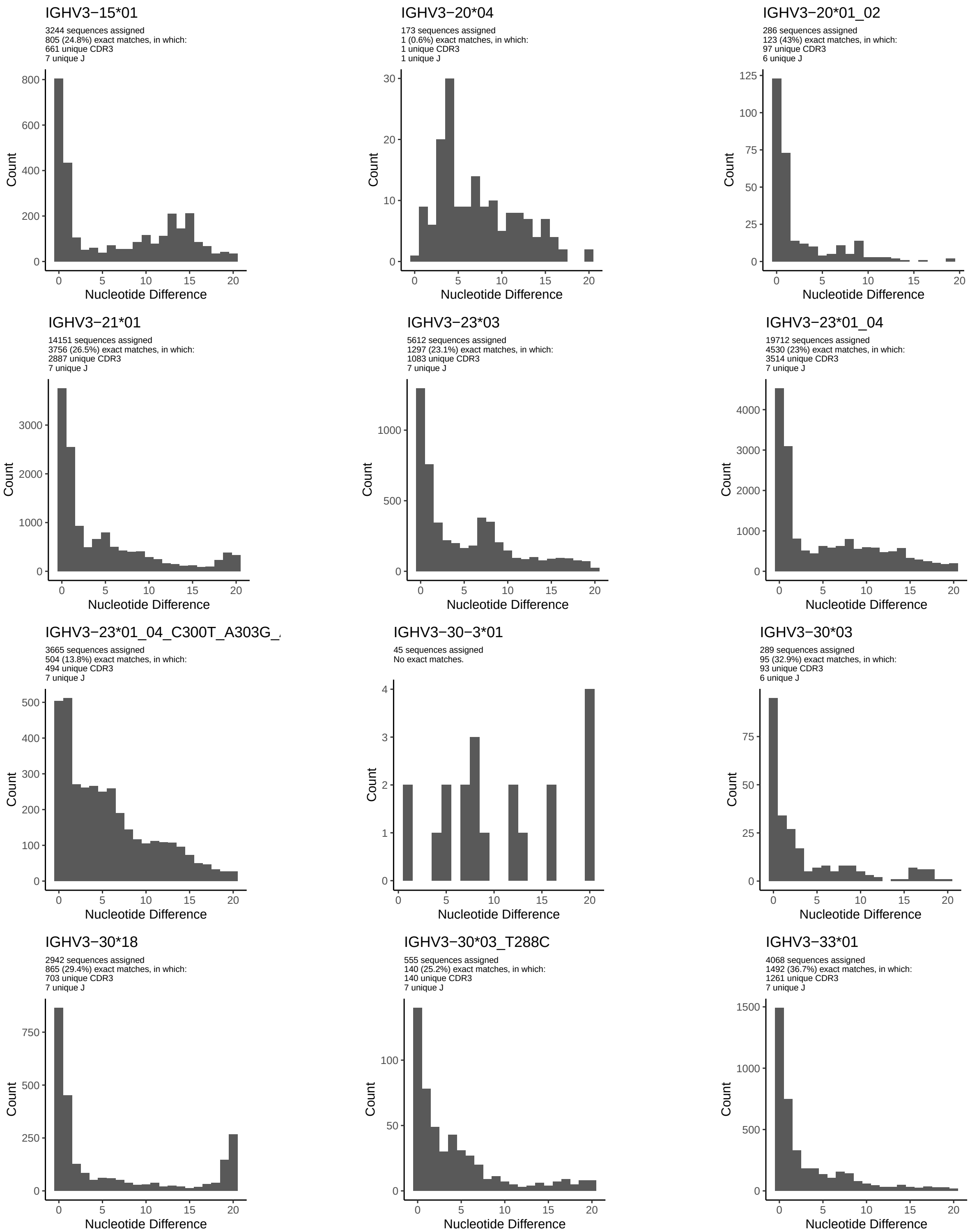


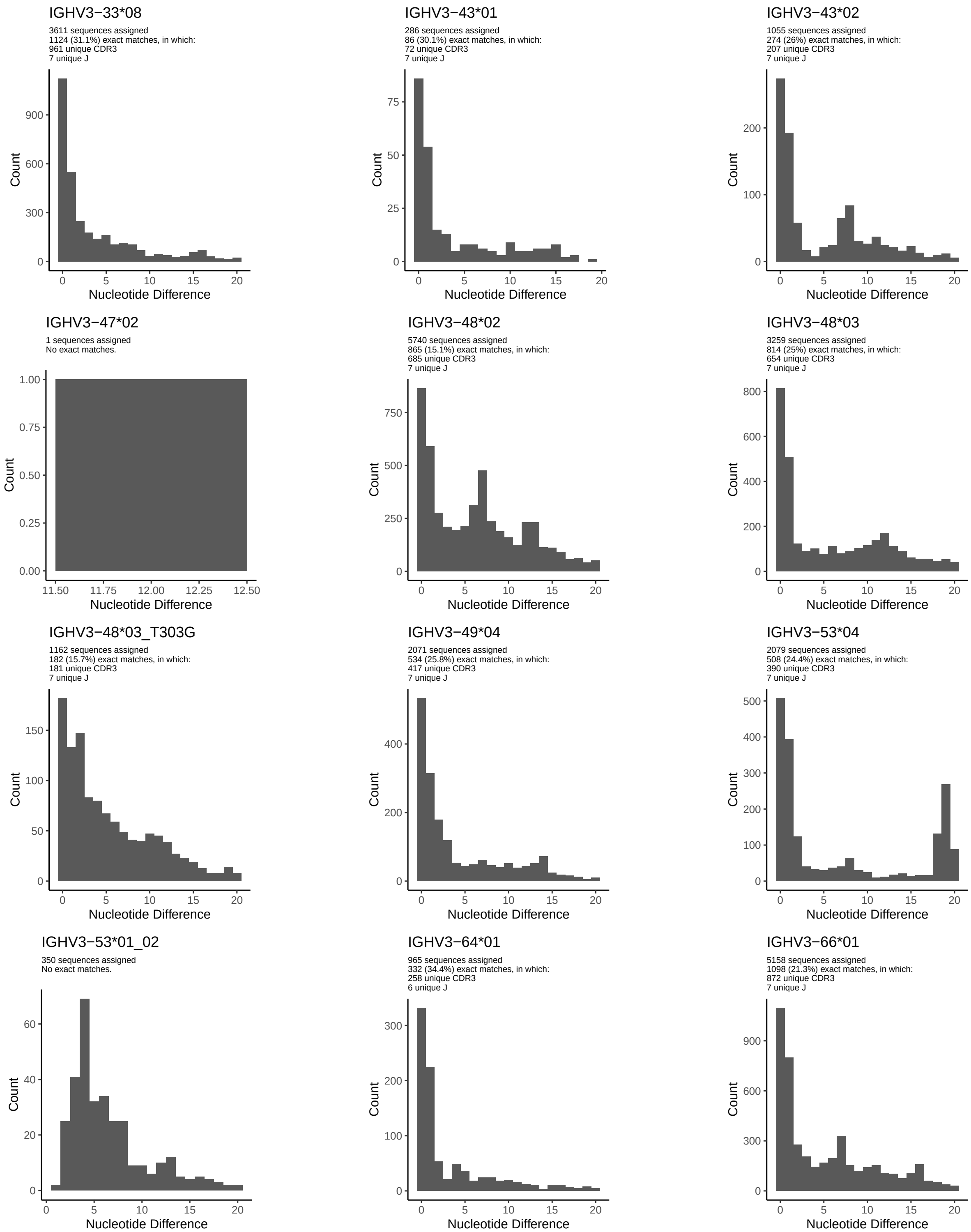

IGHV7−4−1*02_T63C
Count
0
20
40
60
7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 29 30
CDR3 AA Length (unmutated)

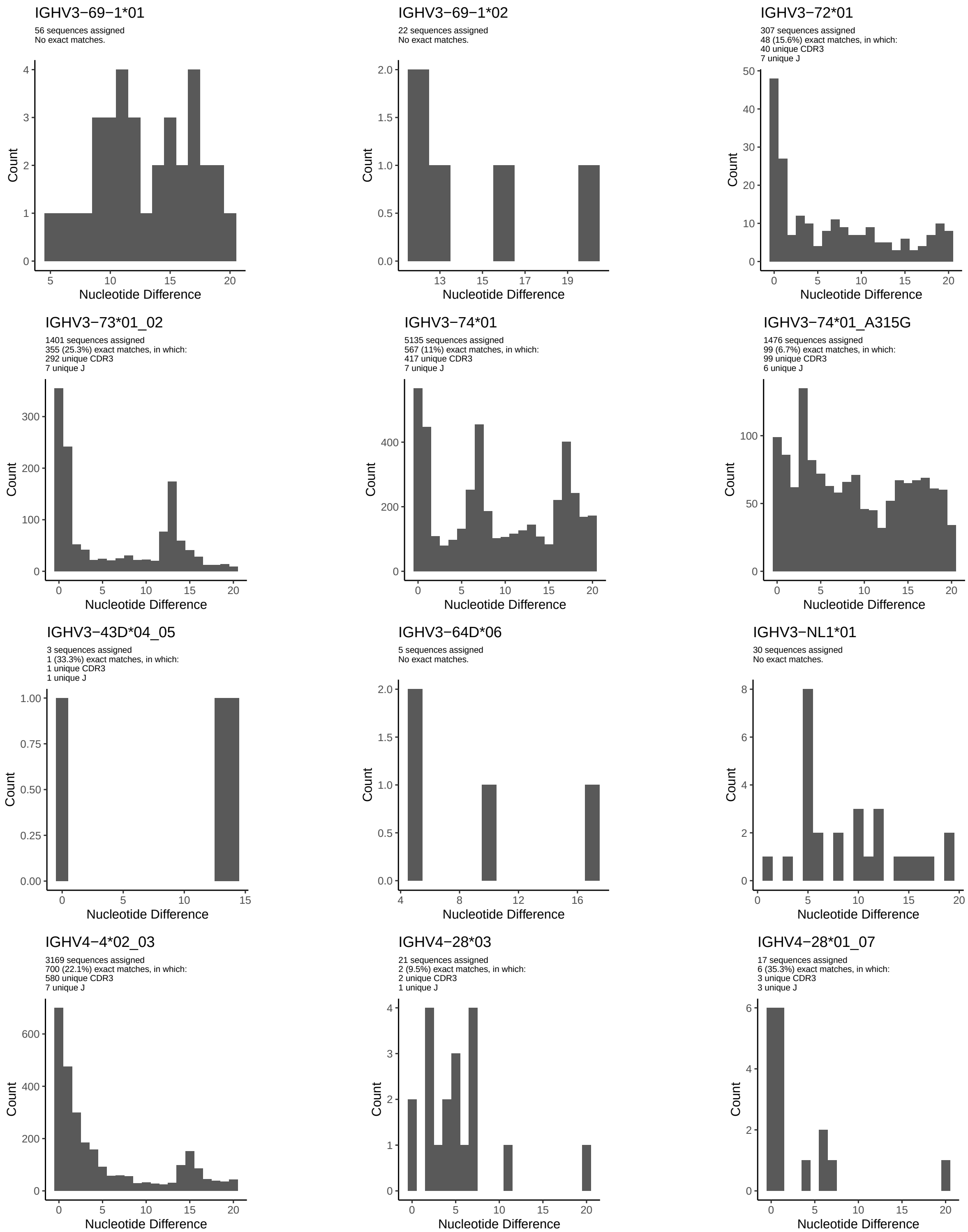
# 2 Variation from germline, in assignments to each allele

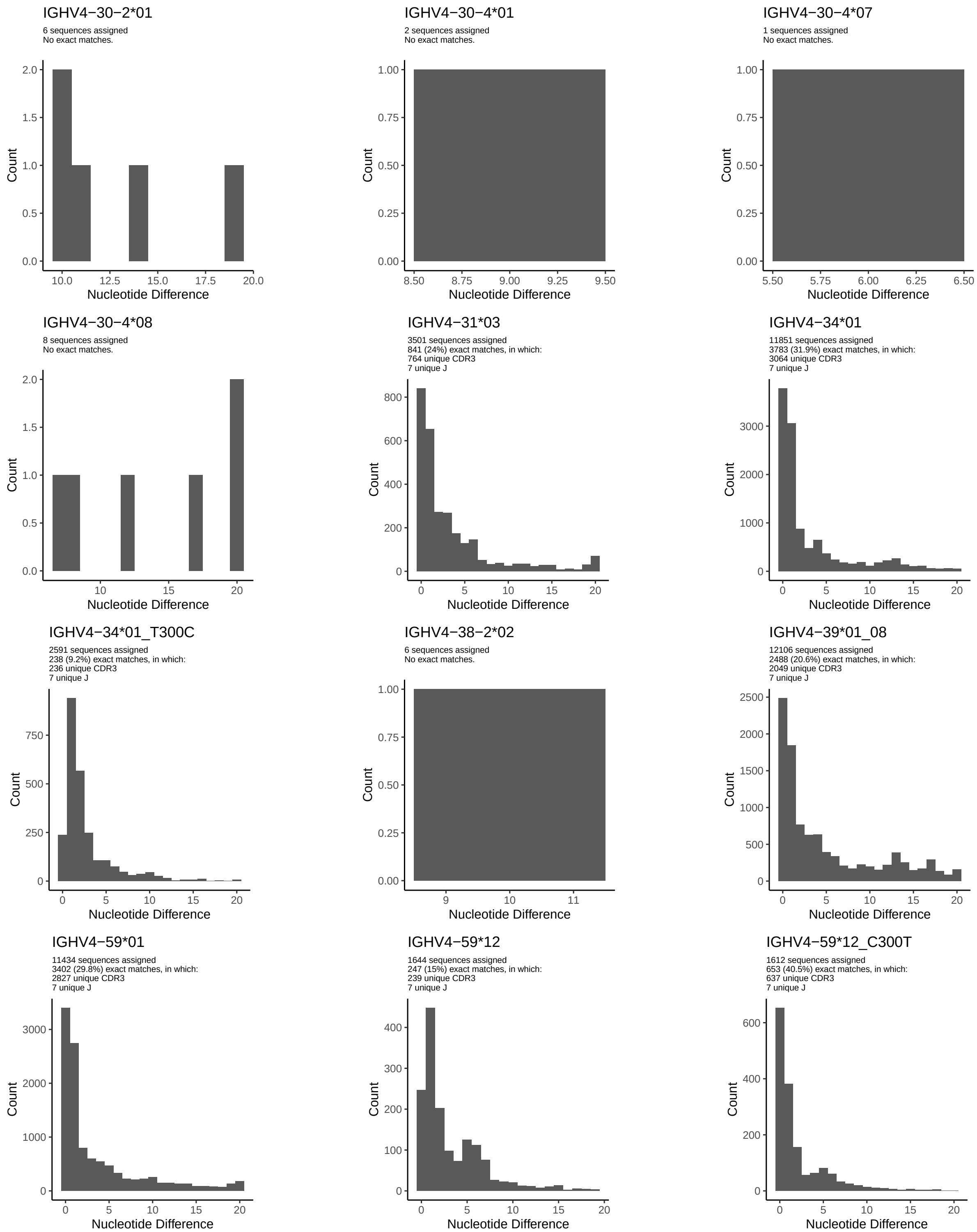



IGHV1−69*02
5387 sequences assigned
2015 (37.4%) exact matches, in which:
1780 unique CDR3
7 unique J
IGHV1−69*04_09
4675 sequences assigned
1606 (34.4%) exact matches, in which:
1400 unique CDR3
7 unique J
IGHV2−5*01
1562 sequences assigned
248 (15.9%) exact matches, in which:
219 unique CDR3
7 unique J
IGHV2−5*02
1468 sequences assigned
273 (18.6%) exact matches, in which:
241 unique CDR3
7 unique J
IGHV2−26*01
293 sequences assigned
121 (41.3%) exact matches, in which:
113 unique CDR3
5 unique J
IGHV2−70*15
1629 sequences assigned
421 (25.8%) exact matches, in which:
377 unique CDR3
6 unique J
IGHV2−70D*04
8 sequences assigned
No exact matches.
IGHV3−7*01
10204 sequences assigned
1886 (18.5%) exact matches, in which:
1530 unique CDR3
7 unique J
IGHV3−9*01
9148 sequences assigned
2155 (23.6%) exact matches, in which:
1690 unique CDR3
7 unique J
IGHV3−11*01
8048 sequences assigned
1557 (19.3%) exact matches, in which:
1248 unique CDR3
7 unique J
IGHV3−13*01
1317 sequences assigned
330 (25.1%) exact matches, in which:
241 unique CDR3
7 unique J
IGHV3−13*06
66 sequences assigned
1 (1.5%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference

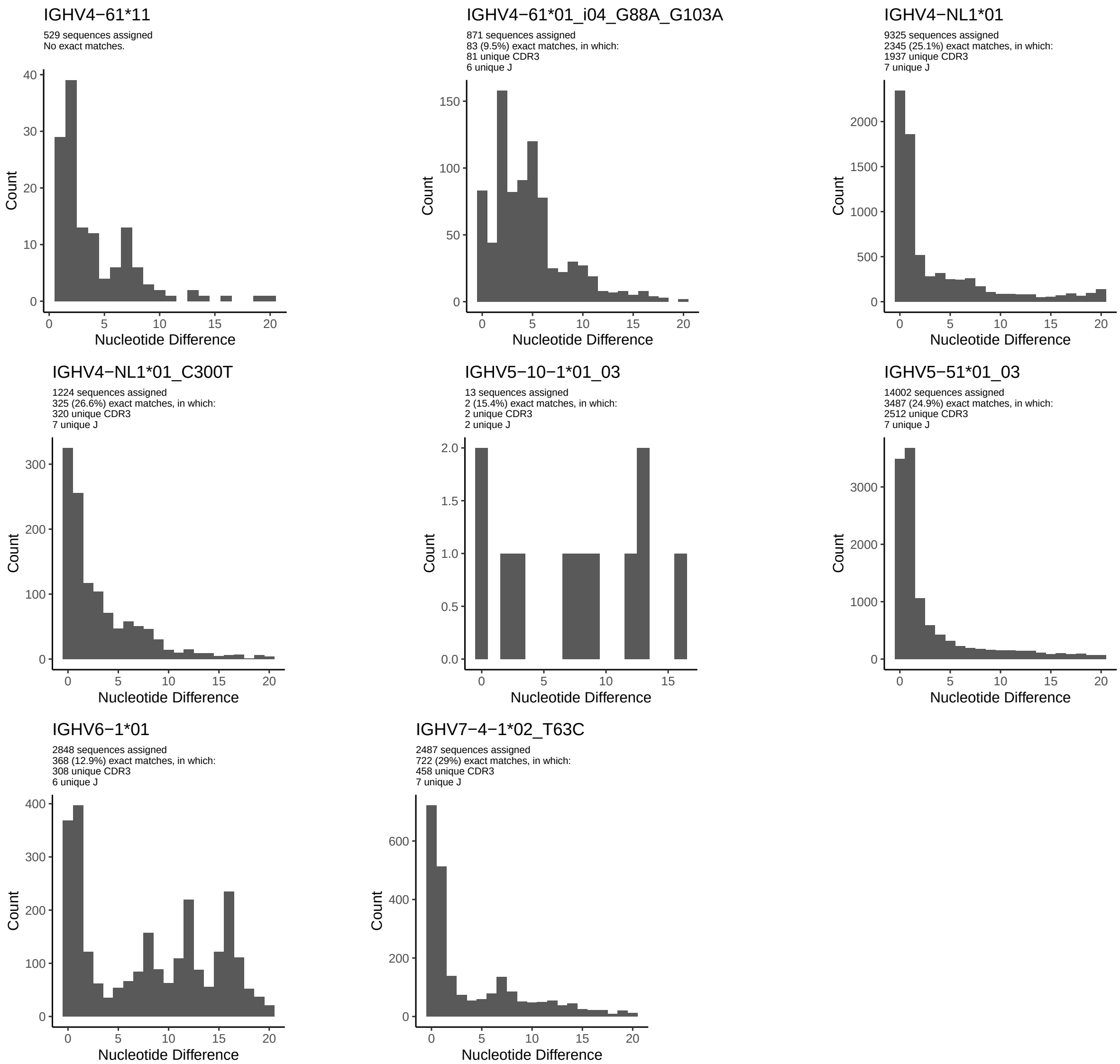

IGHV3−15*01
3244 sequences assigned
805 (24.8%) exact matches, in which:
661 unique CDR3
7 unique J
IGHV3−20*04
173 sequences assigned
1 (0.6%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−20*01_02
286 sequences assigned
123 (43%) exact matches, in which:
97 unique CDR3
6 unique J
IGHV3−21*01
14151 sequences assigned
3756 (26.5%) exact matches, in which:
2887 unique CDR3
7 unique J
IGHV3−23*03
5612 sequences assigned
1297 (23.1%) exact matches, in which:
1083 unique CDR3
7 unique J
IGHV3−23*01_04
19712 sequences assigned
4530 (23%) exact matches, in which:
3514 unique CDR3
7 unique J
IGHV3−23*01_04_C300T_A303G_
3665 sequences assigned
504 (13.8%) exact matches, in which:
494 unique CDR3
7 unique J
IGHV3−30−3*01
45 sequences assigned
No exact matches.
IGHV3−30*03
289 sequences assigned
95 (32.9%) exact matches, in which:
93 unique CDR3
6 unique J
IGHV3−30*18
2942 sequences assigned
865 (29.4%) exact matches, in which:
703 unique CDR3
7 unique J
IGHV3−30*03_T288C
555 sequences assigned
140 (25.2%) exact matches, in which:
140 unique CDR3
7 unique J
IGHV3−33*01
4068 sequences assigned
1492 (36.7%) exact matches, in which:
1261 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV3−33*08
3611 sequences assigned
1124 (31.1%) exact matches, in which:
961 unique CDR3
7 unique J
IGHV3−43*01
286 sequences assigned
86 (30.1%) exact matches, in which:
72 unique CDR3
7 unique J
IGHV3−43*02
1055 sequences assigned
274 (26%) exact matches, in which:
207 unique CDR3
7 unique J
IGHV3−47*02
1 sequences assigned
No exact matches.
IGHV3−48*02
5740 sequences assigned
865 (15.1%) exact matches, in which:
685 unique CDR3
7 unique J
IGHV3−48*03
3259 sequences assigned
814 (25%) exact matches, in which:
654 unique CDR3
7 unique J
IGHV3−48*03_T303G
1162 sequences assigned
182 (15.7%) exact matches, in which:
181 unique CDR3
7 unique J
IGHV3−49*04
2071 sequences assigned
534 (25.8%) exact matches, in which:
417 unique CDR3
7 unique J
IGHV3−53*04
2079 sequences assigned
508 (24.4%) exact matches, in which:
390 unique CDR3
7 unique J
IGHV3−53*01_02
350 sequences assigned
No exact matches.
IGHV3−64*01
965 sequences assigned
332 (34.4%) exact matches, in which:
258 unique CDR3
6 unique J
IGHV3−66*01
5158 sequences assigned
1098 (21.3%) exact matches, in which:
872 unique CDR3
7 unique J
Count
Nucleotide Difference

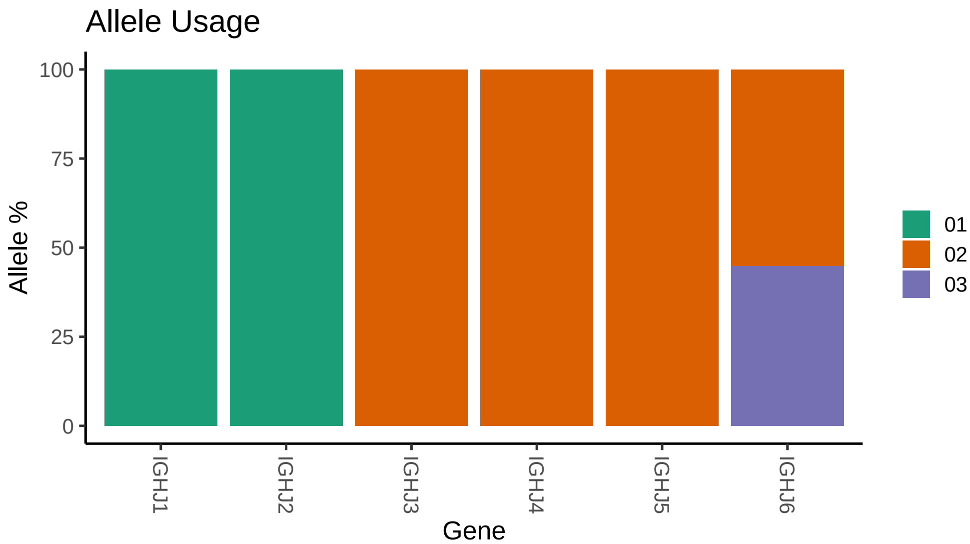

IGHV3−69−1*01
56 sequences assigned
No exact matches.
IGHV3−69−1*02
22 sequences assigned
No exact matches.
IGHV3−72*01
307 sequences assigned
48 (15.6%) exact matches, in which:
40 unique CDR3
7 unique J
IGHV3−73*01_02
1401 sequences assigned
355 (25.3%) exact matches, in which:
292 unique CDR3
7 unique J
IGHV3−74*01
5135 sequences assigned
567 (11%) exact matches, in which:
417 unique CDR3
7 unique J
IGHV3−74*01_A315G
1476 sequences assigned
99 (6.7%) exact matches, in which:
99 unique CDR3
6 unique J
IGHV3−43D*04_05
3 sequences assigned
1 (33.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−64D*06
5 sequences assigned
No exact matches.
IGHV3−NL1*01
30 sequences assigned
No exact matches.
IGHV4−4*02_03
3169 sequences assigned
700 (22.1%) exact matches, in which:
580 unique CDR3
7 unique J
IGHV4−28*03
21 sequences assigned
2 (9.5%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV4−28*01_07
17 sequences assigned
6 (35.3%) exact matches, in which:
3 unique CDR3
3 unique J
Count
Nucleotide Difference



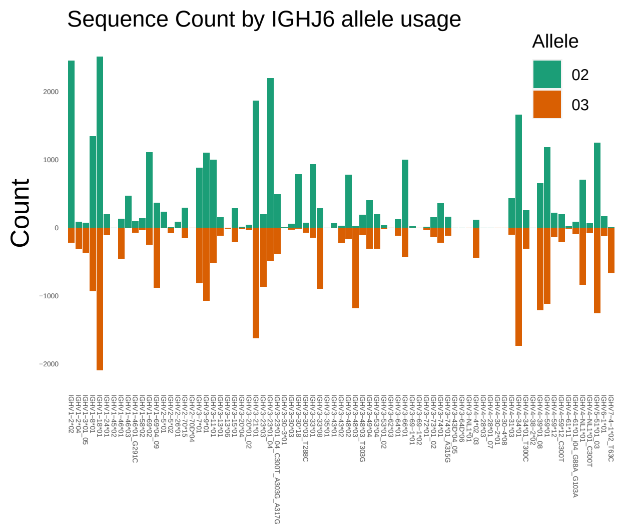
IGHV4−30−2*01
6 sequences assigned
No exact matches.
IGHV4−30−4*01
2 sequences assigned
No exact matches.
IGHV4−30−4*07
1 sequences assigned
No exact matches.
IGHV4−30−4*08
8 sequences assigned
No exact matches.
IGHV4−31*03
3501 sequences assigned
841 (24%) exact matches, in which:
764 unique CDR3
7 unique J
IGHV4−34*01
11851 sequences assigned
3783 (31.9%) exact matches, in which:
3064 unique CDR3
7 unique J
IGHV4−34*01_T300C
2591 sequences assigned
238 (9.2%) exact matches, in which:
236 unique CDR3
7 unique J
IGHV4−38−2*02
6 sequences assigned
No exact matches.
IGHV4−39*01_08
12106 sequences assigned
2488 (20.6%) exact matches, in which:
2049 unique CDR3
7 unique J
IGHV4−59*01
11434 sequences assigned
3402 (29.8%) exact matches, in which:
2827 unique CDR3
7 unique J
IGHV4−59*12
1644 sequences assigned
247 (15%) exact matches, in which:
239 unique CDR3
7 unique J
IGHV4−59*12_C300T
1612 sequences assigned
653 (40.5%) exact matches, in which:
637 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−61*11
529 sequences assigned
No exact matches.
IGHV4−61*01_i04_G88A_G103A
871 sequences assigned
83 (9.5%) exact matches, in which:
81 unique CDR3
6 unique J
IGHV4−NL1*01
9325 sequences assigned
2345 (25.1%) exact matches, in which:
1937 unique CDR3
7 unique J
IGHV4−NL1*01_C300T
1224 sequences assigned
325 (26.6%) exact matches, in which:
320 unique CDR3
7 unique J
IGHV5−10−1*01_03
13 sequences assigned
2 (15.4%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV5−51*01_03
14002 sequences assigned
3487 (24.9%) exact matches, in which:
2512 unique CDR3
7 unique J
IGHV6−1*01
2848 sequences assigned
368 (12.9%) exact matches, in which:
308 unique CDR3
6 unique J
IGHV7−4−1*02_T63C
2487 sequences assigned
722 (29%) exact matches, in which:
458 unique CDR3
7 unique J
Count
Nucleotide Difference

# 3 Allele usage in potential haplotype anchor genes



# 4 Haplotype plots



# 5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 110/M64 110/M64 110/M64 110/53/99f06df3c81203278
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 110/M64 110/M64 110/M64 110/53/99f06df3c8
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 110/M64 110/M64 110/M64 110/53/99f06df3c812032
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 46 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 46 01_G291C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
```